

Ph #11:-

Computer misuse & Criminal law

Heft
ex much
small

Goal is to punish
↑ focus on action
harmful for
society
as a whole

Computing & Criminal law:

→ Modern business process is done through utilizing Computer Software and hardware. Some form of Computer system is used in it.

→ There has been a sharp rise in the number of crimes involving Computing; and the Internet has undoubtedly created new security risks.

→ In recent years, people (especially the media) have been more worried about internet misuse than general computer misuse. However, crimes involving computers are a big part of "white-collar crime" (Non-violent crimes committed for financial gain).

Reform of the criminal law:

Evolving challenges: The rapid growth of technology brought new types of crimes that existing laws could not effectively address, such as hacking and unauthorized access.

The Computer Misuse Act 1990

⇒ The Computer Misuse Act 1990 was introduced in the UK to address crimes involving the misuse of computers such as hacking or spreading viruses.

⇒ Before this law it was difficult to convict hackers because existing laws were not specific enough.

Main offenses in the Act

1. Unauthorized Access to a Computer (Section-1)

- It is illegal to access a computer without permission.
- This means causing a computer to perform tasks (like opening a program or viewing data) when you know you don't have the right to do so.

◦ Ex:

Logging to someone else's email account without their consent.

Penalty: Up to £5,000 fine or six months in prison.

2. Unauthorized Access with intent to Commit a Serious Crime (Section-1)

◦ If you access a computer without permission intending to commit a bigger crime, it's a more serious offense.

ex:

- A hacker accessing medical records to blackmail someone.
- A terrorist trying to manipulate an air traffic control system.

Penalty: Up to five years in prison or an unlimited fine.

3. Unauthorized Modification of Computer Data

It is illegal to make unauthorized changes to computer data or programs with harmful intent.

Example:

- Spreading a virus or worm.
- Encrypting company files and demanding payment to unblock them (ransomware).
- Changing browser settings secretly (redirection home pages).

- Installing programs that dial expensive phone call (Premium rate dialers).

Penalty: up to five years in prison or an unlimited fine.

Why this Act important?

- It ensures that actions like hacking, spreading viruses and tampering with data are treated as crimes.
- The law applies even if the computer or offender is located outside the UK.

2004 Review of the Computer Misuse Act:

→ The All-party Parliamentary Internet Group reviewed the Computer Misuse Act 1990 and recommended minor updates to improve its effectiveness:

1. New offence for Denial of service (DoS) Attack:

Add a law to sue individuals impairing access to data, especially for DoS attacks.

2. Stronger Penalties for Hacking:

Increase penalties from 6-months to 2-Years.

3. Few Prosecutions:

Only 32 were brought between 1999-2000, mainly due to lack of awareness about the Act's scope and limited police resources.

4. Recommendations:

- o Raise public awareness of the Act.
- o Provide more resources to police for investigating cybercrime.

Categories of Misuse

→ These are different ways people can misuse computers. These actions are considered illegal and can have serious consequences.

→ These are various categories of computer misuse:

1. Computer fraud.
2. Unauthorized obtaining of information from a computer.

- Computer hacking
 - Eavesdropping on computer
 - Making unauthorized use of computers for personal benefit.
3. Unauthorized alteration or destruction of information stored on a computer.
 4. Denying access to an authorized user.
 5. The unauthorized removal of information stored on a computer.

Computer fraud

Illegal activity that involves the use of computers.

- When someone uses a computer or the internet to trick or deceive someone else in order to gain something of value or to cause harm.
- it is a conduct which involves manipulation of a computer, by whatever method, in order dishonestly to obtain money, property or some other advantage of value.
- These are three main types of computer fraud:

1. Input frauds:

When someone intentionally enters false or misleading information/data into a computer. **ex:** Change a customer's address so that their order is sent to different location.

2. Output frauds:

When someone steals or misuses information that comes out of a computer system.

ex: They might steal a list of customer names and addresses to sell to a marketing company.

3. Program frauds: When someone changes a computer program to do something dishonest.

ex: might change a program so that it calculates their salary incorrectly, giving them more money than they deserve.

Unauthorized obtaining of information

The second form of misuse identified by the Law Commission was unauthorized obtaining of information from a computer.

→ It is subdivided as :

1. Computer Hacking
2. Eavesdropping on Computers
3. Unauthorized use of Computers for personal benefit.

1. **Computer Hacking:** - involves black hat hacking.

- Accessing a computer without the authorization of the owner. In this case the person accesses the computer secretly for stealing information, data or manipulation of data for diverse purposes.

- Prior to 1990, when the computer misuse act was enacted, it was difficult to convict anyone of computer hacking.

2. **Eavesdropping:**

- it involves secret listening or watching someone's computer activity without their permission.

- This can be done by using software to intercept communications, or by physically accessing someone's computer.

Physical - eavesdropping:

Digital eavesdropping:

Network sniffing (intercepting & analyzing network traffic)

Man-in-the-middle attack

Keylogging -> phishing

Listening devices (microphone, or hidden cameras)

Visual surveillance (watching conversation through windows, keyholes, or hidden cameras)

Eavesdropping is a serious privacy violation, in many jurisdictions, eavesdropping is a crime.

3. Unauthorized use of Computers for Personal Benefit:

→ This category includes any activity that involves using someone else's computer for own personal benefit without their permission. This can include:

- Using someone else's computer to play games or watch videos.
- Using someone else's computer to access the internet.
- Using someone else's computer to send or receive email.
- Using someone else's computer to send or store data.

→ Unauthorized use of computers can be a serious offense, & it can lead to criminal charges.

Unauthorized alteration or destruction of information

→ Computers are valuable because they can store and make large amounts of data easily accessible.

→ However, this capability raises concerns about the security of that data.

→ People who intentionally destroy or alter data without permission can face criminal charges.

→ Two main laws address this issue:

- Criminal damage laws

- Computer Misuse Act

★ Unauthorized removal of information stored on a computer